

WEEK 4 PROJECT SUMMARY

The project focuses on understanding appointment attendance patterns and identifying factors associated with missed appointments. HealthConnect is a fictional healthcare provider facing challenges such as patient no-shows, inefficient use of appointment slots, and difficulty understanding the reasons behind missed appointments. The main objective is to use data to support better appointment management and patient engagement.

For this project, I reviewed the HealthConnect Appointment Dataset, Data Dictionary, and Clinic Knowledge Base. The HealthConnect Appointment Dataset is a fictional and anonymized dataset containing appointment records with information about patient demographics, appointment details, booking behaviour, previous appointment history, previous no-shows, reminder information, distance to the clinic, waiting time, and appointment outcomes. The main outcome variable is appointment_outcome, which contains three categories: Attended, No-Show, and Cancelled.

KEY WEEK 4 OBSERVATIONS

The initial review identified several variables that can support the investigation of appointment attendance and no-shows. These include age, age group, gender, appointment type, appointment day, appointment time, booking lead days, previous appointments, previous no-shows, reminder sent, reminder channel, distance to clinic, and waiting time.

The initial data-quality assessment also identified areas requiring validation, including duplicate appointment IDs, missing distance and waiting-time information, consistency between previous appointments and previous no-shows, reminder-field consistency, date consistency, and correct classification of cancelled appointments. Cancelled appointments will be kept separate from no-shows to avoid misleading results.

PROPOSED ANALYSIS

The analysis will follow a structured process of data understanding, data-quality assessment, data preparation, exploratory analysis, KPI analysis, and insight development.

Five potential KPIs have been identified for subsequent stages No-Show Rate, Attendance Rate, Cancellation Rate, Reminder Coverage Rate, Repeat No-Show Rate

These KPIs will help measure the scale of missed appointments and investigate factors that may be associated with attendance.

The analysis will also examine business questions relating to patient characteristics, appointment scheduling, previous no-show behaviour, reminder usage, reminder channels, booking lead time, and distance to the clinic.

MICROSOFT POWER BI

For this project, I will use Microsoft Power BI as the main data analytics and visualization tool. Power BI will be used to import the HealthConnect Appointment Dataset, clean and transform the data, analyze appointment outcomes, calculate the proposed KPIs, identify patterns in no-shows and attendance, and create interactive dashboards and visualizations.

Using Power BI will allow the appointment data to be transformed into clear and interactive insights. The dashboards will help examine how different factors, such as age group, appointment type, previous no-shows, reminders, booking lead time, and distance to the clinic, relate to appointment outcomes.

KEY CONSIDERATIONS AND WEEK 5 FOCUS

The project is subject to limitations such as missing information, limited available variables, and the fictional nature of the dataset. There is also a risk of interpreting associations as causal relationships. Patient information must be handled responsibly, with analysis focused on identifying patterns rather than blaming or labelling individuals.

Week 5 will focus on exploratory data analysis using the HealthConnect Appointment Dataset in Microsoft Power BI. This will include importing and preparing the dataset, performing data-quality checks, calculating the proposed KPIs, examining no-show and attendance patterns across relevant variables, and developing interactive Power BI visualizations and dashboards.

The findings from the analysis will be used to develop evidence-based insights and recommendations that can help HealthConnect improve appointment attendance, reduce missed appointments, use appointment slots more efficiently, and improve the overall patient support experience.